

Spirit . Intellect . Purpose



**University of
Batangas**

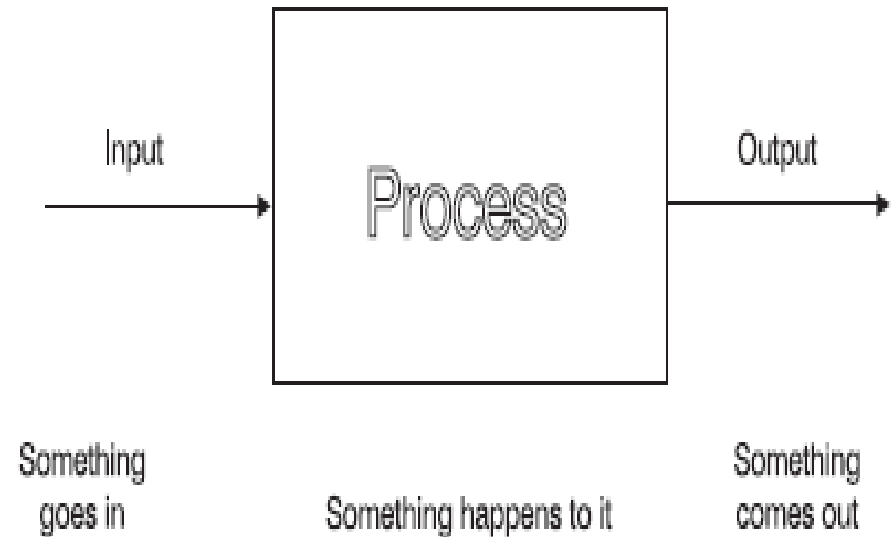


Microprocessor System

Introduction



- The word 'system' is used to describe any organization or device that includes three features.
- A system must have at least one input, one output and must do something, i.e. it must contain a process.
- Often there are many inputs and outputs. Some of the outputs are required and some are waste products. To a greater or lesser extent, all processes generate some waste heat.



Example

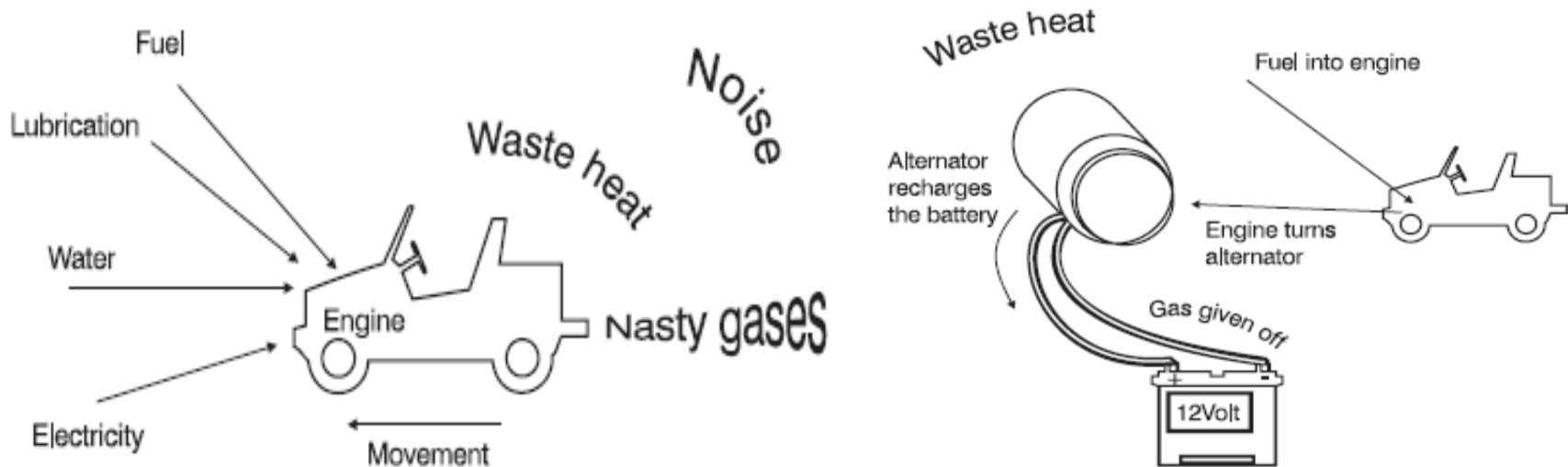


A motor car will usually require fuel, water for cooling purposes and a battery to start the engine and provide for the lights and instruments.

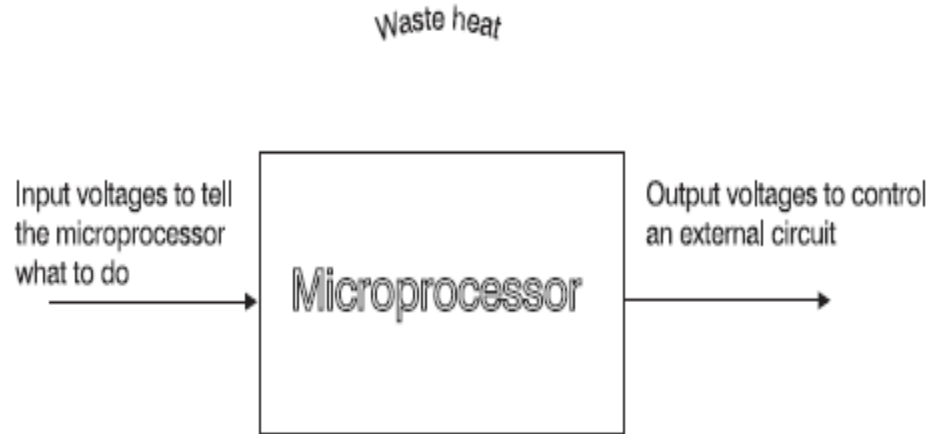
Its process is to burn the fuel and extract the energy to provide transportation for people and goods.

The outputs are the wanted movement and the unwanted pollutants such as gases, heat, water vapour and noise.

The motor car contains other systems within it. An electricity is required as an input to start the engine and provide the lights



A microprocessor system



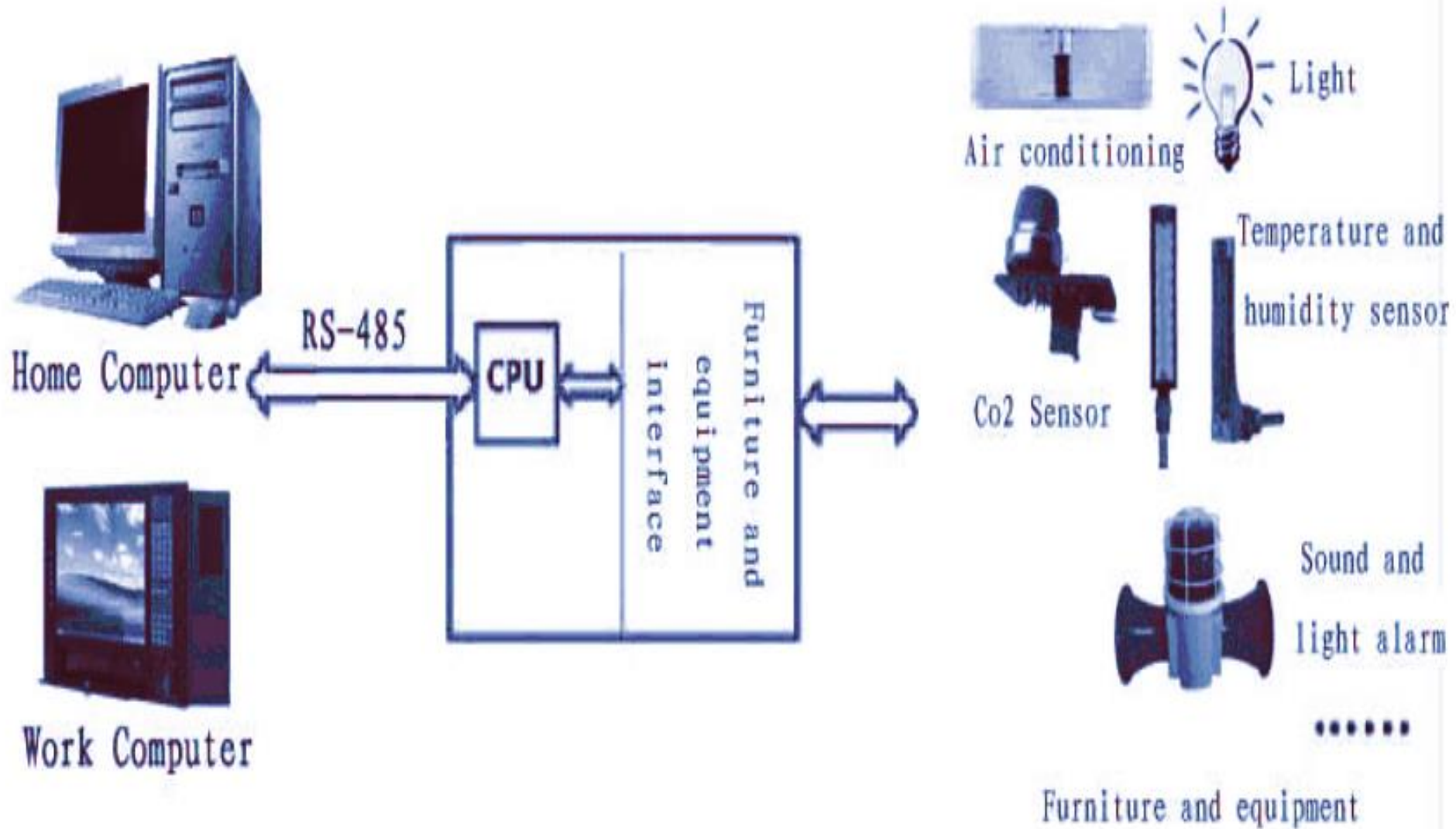
Like any other system, a microprocessor has inputs, outputs and a process.

The inputs and outputs of a microprocessor are a series of voltages that can be used to control external devices.

The process involves analysing the input voltages and using them to 'decide' on the required output voltages.

The decision is based on previously entered instructions that are followed quite blindly, sensible or not.

Example: Microprocessor Based System



Terminology



- **Microprocessor-based system**

This is any system that contains a microprocessor, and does not necessarily have anything to do with computing. In fact, despite all the hype, computers use only a small proportion of all the microprocessors manufactured.

- **Microcomputer**

The particular microprocessor-based systems that happen to be used as a computer are called microcomputers. The additional circuits required for a computer can be built into the same integrated circuit giving rise to a single chip microcomputer.

Microcomputers are designed to be used by individuals, whether in the form of PCs, workstations or notebook computers.

A microcomputer contains a CPU on a microchip (the microprocessor), a memory system (typically ROM and RAM), a bus system and I/O ports, typically housed in a motherboard.

Terminology



- **Microcontroller**

This is a complete microprocessor-based control system built onto a single chip. It is small and convenient but doesn't do anything that could not be done with a microprocessor and a few additional components.

Integrated circuits

An electronic circuit fabricated out of a solid block of semiconductor material. This design of circuit, often called a solid state circuit, allows for very complex circuits to be constructed in a small volume. An integrated circuit is also called a 'chip'.

What is microprocessor?



- A microprocessor is an integrated circuit that contains the functions of a central processing unit (CPU) of a computer.

Importance: Essential for the operation of computers and many other electronic devices.

Microprocessor



- μ P is a complex, powerful device:
 - Able to process huge amounts of data.
 - Built using transistors on silicon die.
 - Needs external components to support operation.
- Used in wide variety of applications.
 - Take advantage of processing power.
- Microcomputer system – support μ P operations.

Microprocessor



- Microprocessors also control the logic of almost all digital devices, from clock radios to fuel-injection systems for automobiles.
- Three basic characteristics differentiate microprocessors:

Instruction set: The set of instructions that the microprocessor can execute.

Bus width : The number of bits processed in a single instruction.

Clock speed : Given in hertz (Hz), the clock speed determines how many instructions per second the processor can execute.

-

32-Bit Versus 64-Bit CPU



- A **32-bit processor** means it can perform operations on 32-bit data; therefore, the size of registers is 32 bits and ALU also performs 32-bit operations.
- A **64-bit CPU** performs operation in 64-bit data; therefore, it contains 64-bit register and 64-bit ALU.
- Most desktop and server computers are using AMD and Intel processors; they might use 32 bits or 64 bits. Intel and AMD processor use the same architecture; this means a program in computer with Intel processor can run on a computer with AMD processor.

History of Computer



- Has undergone significant improvements:
 - 5 generations until now.
 - Tied to development of electronics, semiconductors.
- What's next?
 - Conventional computing:
 - Advancements in semiconductor technology.
 - Smaller, faster, less power.
 - Unconventional computing:
 - Quantum computer.
 - Chemical computer.
 - Molecular computer.

History of Computer



Generation	Period	Technology Used
1st Gen.	1946-1959	Vacuum Tubes
2nd Gen.	1959-1965	Transistors
3rd Gen.	1965-1971	Integrated Circuits
4th Gen.	1971-1980	Microprocessors
5th Gen.	1980-present	AI & ULSI

Generations of Computer

First Generation



- Vacuum Tubes as switches.
- Magnetic drums as memory.
- Very big, unreliable, slow.
- ENIAC (Electronic Numerical Integrator And Computer),
UNIVAC (UNIVersal Automatic Computer).



Electronic Numerical Integrator and Computer (ENIAC)



Vacuum Tubes

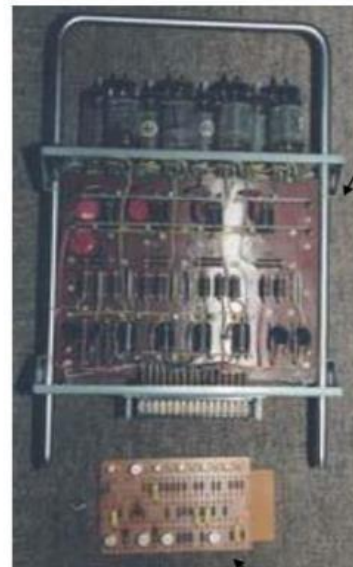
Second Generation



- ❑ After invention of **transistors**.
- ❑ Smaller, faster, cheaper.
- ❑ Limited to military and business use.

Improved programming languages (e.g., COBOL, FORTRAN).

• **Examples:** IBM 1401, CDC 1604.



Vacuum tube circuit

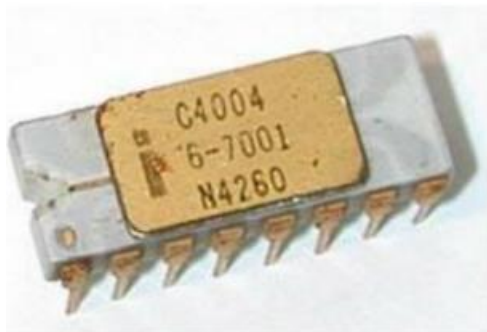
Transistor circuit



Third Generation



- After invention of **Integrated Circuits (IC)**.
 - Many transistors can be packed into IC.
- Intel 8008, Intel 4004.
- Medium Scale Integration (MSI) and Large Scale Integration (LSI).



Early Intel Microprocessor

Third Generation



Third Generation Computers



PC



Laptop

Fourth Generation



- Improvements in IC technology, μ P design.
- More transistors → more processing power.
- Very Large Scale Integration (VLSI).
 - Intel Montecito Itanium: 1 bln. transistors.
- 64-Bit microprocessors.



Fifth Generation



- Fifth-generation computers are available in different sizes and they can now fit on the palm as well.
- Computers have become so fast that they can handle multitasking easily. Also, if the data given by the user is correct, then fifth generation computers are almost 100% accurate in calculations.
- With the use of AI in the fifth generation, computers have become so smart that they can understand human language and recognize pictures of individuals and things. However, all these data are input on the computer or they do so with the help of the internet. They do not have the ability to think as human beings.



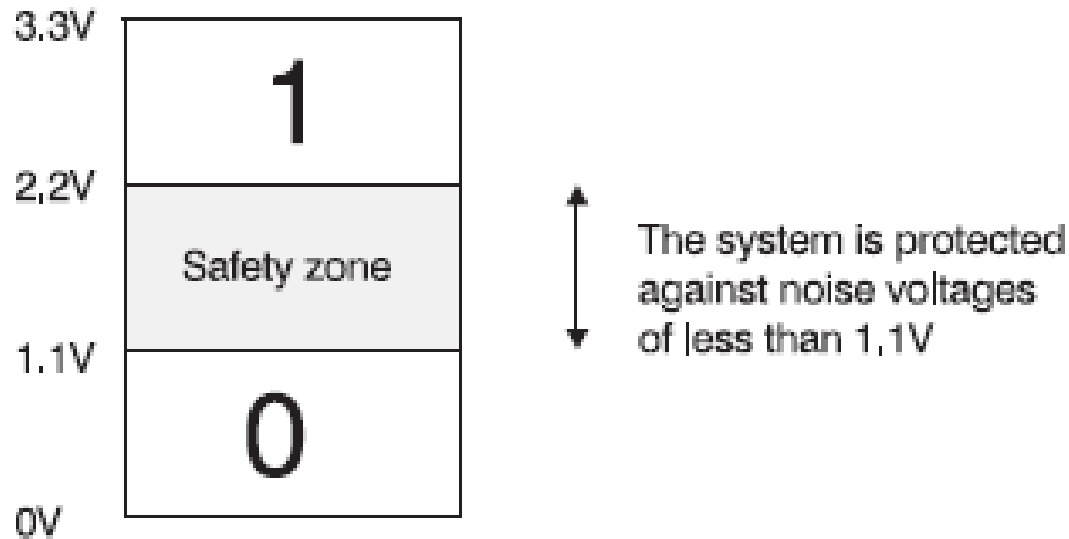
Artificial Intelligence



Binary – the way micros count



- Microprocessors and other digital circuits use only two digits – 0 and 1



The voltages chosen to represent the digits 0 and 1

Bits, bytes and other things



- All the information entering or leaving a microprocessor is in the form of a binary signal, a voltage switching between the two **bit** levels 0 and 1.
- **Nibble** - half a byte.
- **Byte** - simply a collection of 8 bits.
- **Word** - represents 16 bits or **2 bytes**.
- **Double Word** - represents 32 bits or 4 bytes
- **Kilobyte (KB or kbyte)** - is 1024 or 2^{10} bytes
- **Megabyte (MB)** - is a kilokilobyte or 1024 1024 bytes. Numerically this is 2^{20} or 1 048 576 bytes.
- **Gigabyte** - 1024 megabytes which is 2^{30} or 1 073 741 824 bytes
- **Terabyte (TB)** - megamegabyte or 2^{40} or 1 099 511 600 000 bytes

Hexadecimal – the way we communicate with micros



The advantages of hex

- It is very compact. Using a base of 16 means that the number of digits used to represent a given number is usually fewer than in binary.
- It is easy to convert between hex and binary and fairly easy to go with hex

CPU Technology



VON Neumann Computer

CISC – Complex Instruction Set Computer

- - CPU
- - Memory
- - I/O
- - 1 memory module for data and program
- Large variable length instructions
- Multiple addressing mode
- Highly pipelined
- [Pipelining](#) : a technique that allows for simultaneous execution of parts, or stages, of instructions to more efficiently process instructions
 - data and instruction cache for storing the most recent instructions and data
- Instructions that requires multiple nos. of clock cycles for their execution

Harvard Computer (Mark 1)

RISC – Reduced Instruction Set Computer

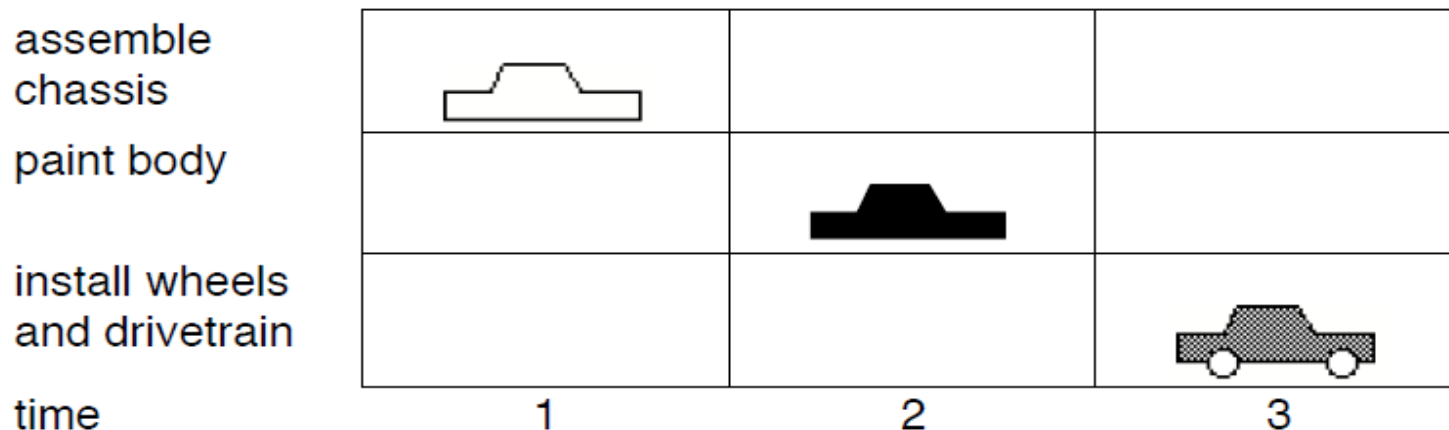
- - CPU
- - Primary Storage
- - Secondary Storage
- - Input / Output
- - 2 memory modules
 - 1 data
 - 1 program
- Reduced variable length Instruction
- Limited addressing mode
- Small number of internal processor, registers
- Not pipelined or less pipelined
- Simple instructions taking 1 clock cycle

Pipelining



A conventional computer executes one instruction at a time with a Program Counter pointing to the instruction currently being executed. Pipelining is analogous to an oil pipeline where the last product may have gone in before the first result comes out. This provides a way to start a task before the first result appears. The computing throughput is now independent of the total processing time.

Conventional processing

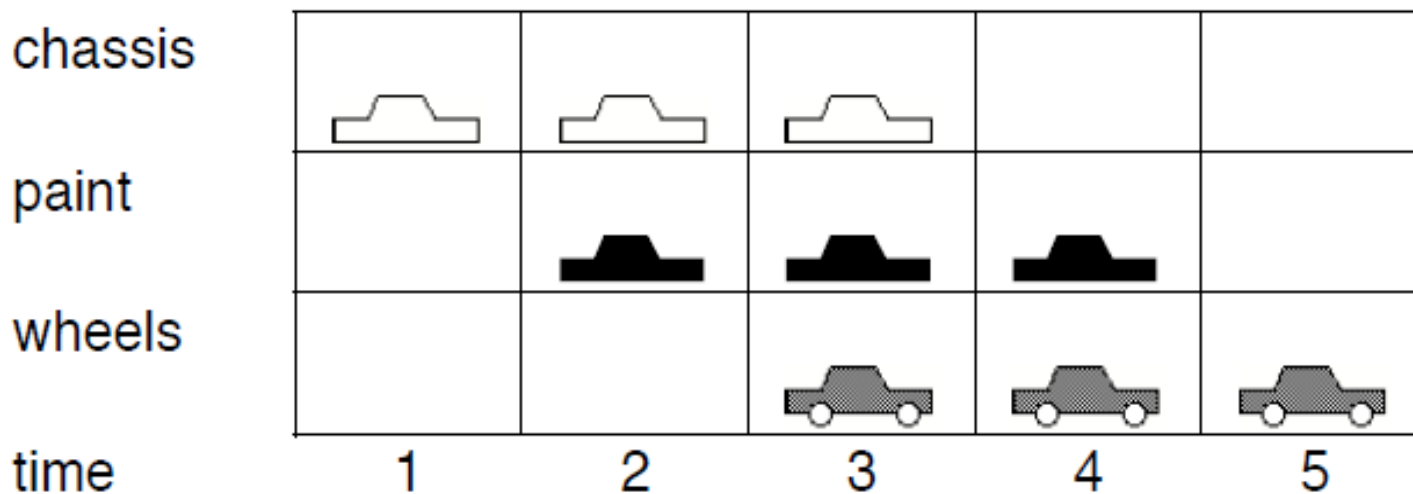


A conventional process would require 9 time units to produce three cars.

Pipelining



Pipelined processing



A pipelined process would require 5 time units to produce the same number of cars.

Four types of processor



1. RISC
2. CISC
3. Hybrid processor – combination of RISC and CISC
4. Special purpose processor – for networking

Types of Automation

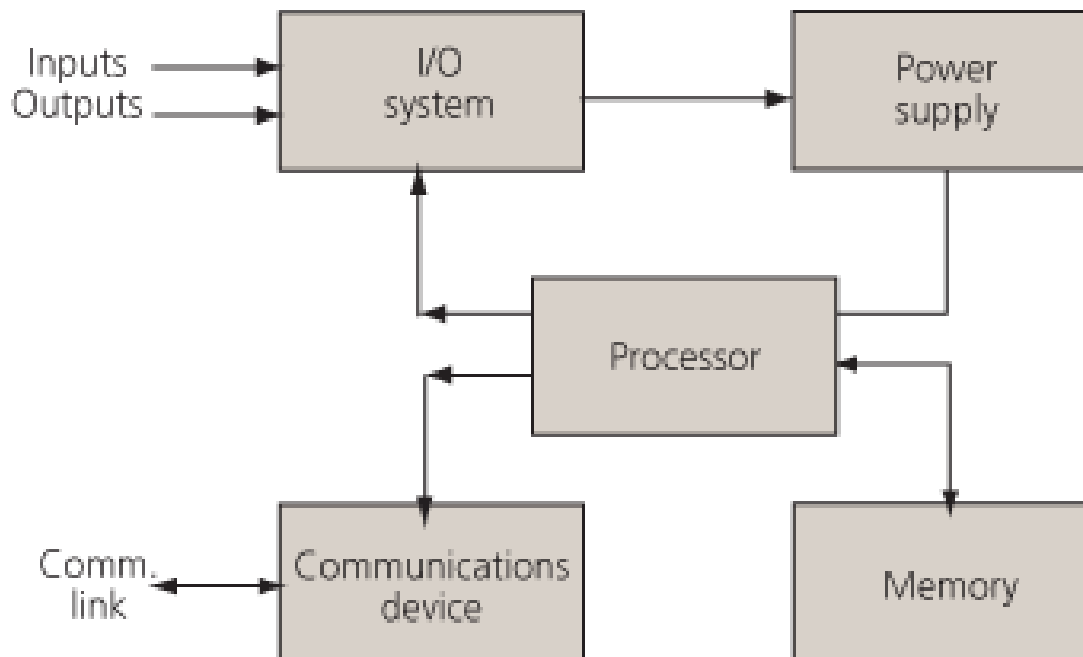


1. **PC based** - good for database ,
interfacing (CISC)
2. **MPU based** – ease to troubleshoot, low
power consumption (CISC)
3. **MCU based** – limited memory, low
power consumption (RISC)
4. **PLC based** – Programmable Logic
Controller. MCU inside (RISC)

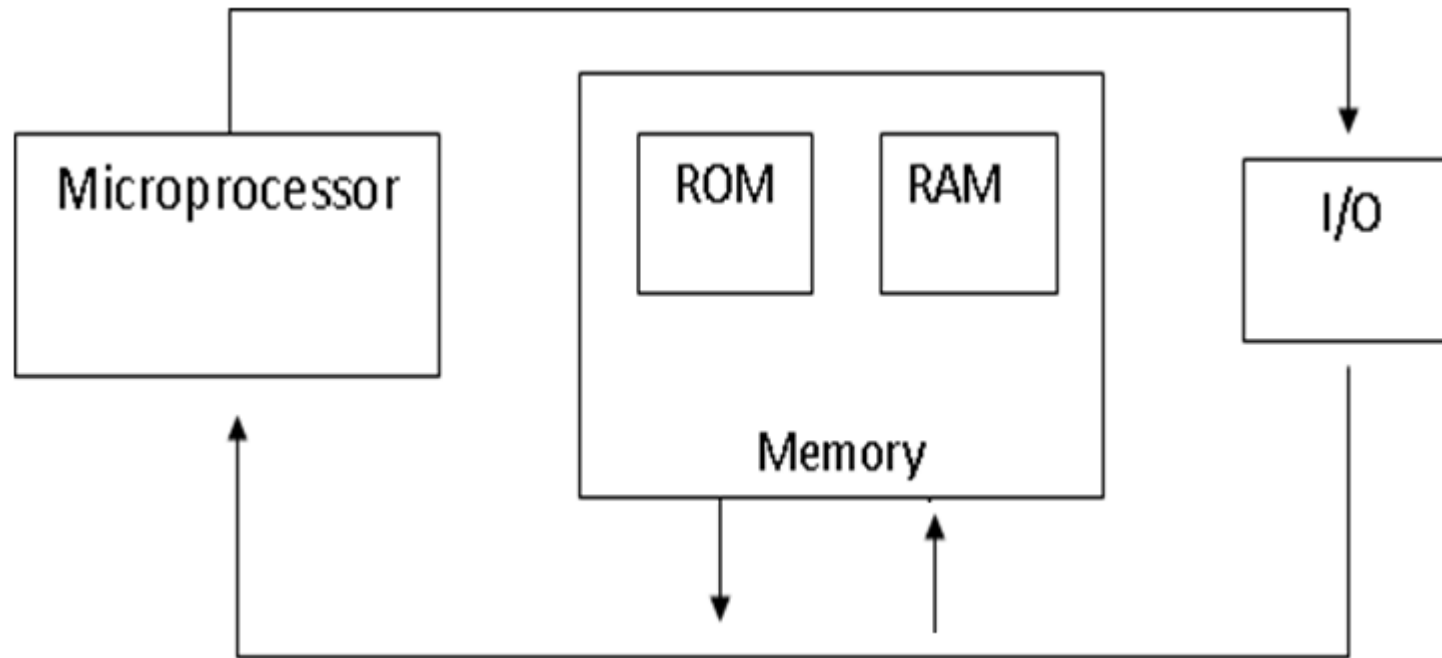
PLC based



A **programmable logic controller (PLC)** or **programmable controller** is a digital computer used for automation of electromechanical processes, such as control of machinery on factory assembly lines, amusement rides, or lighting fixtures. PLCs are used in many industries and machines, such as packaging and semiconductor machines.



Microprocessor



What are registers for?



- storage areas inside the microprocessor
- store the data that is going to be used, they store the instructions that are to be used and they store the results obtained.
- involve tri-state buffers to control the direction of data flow.

Registers



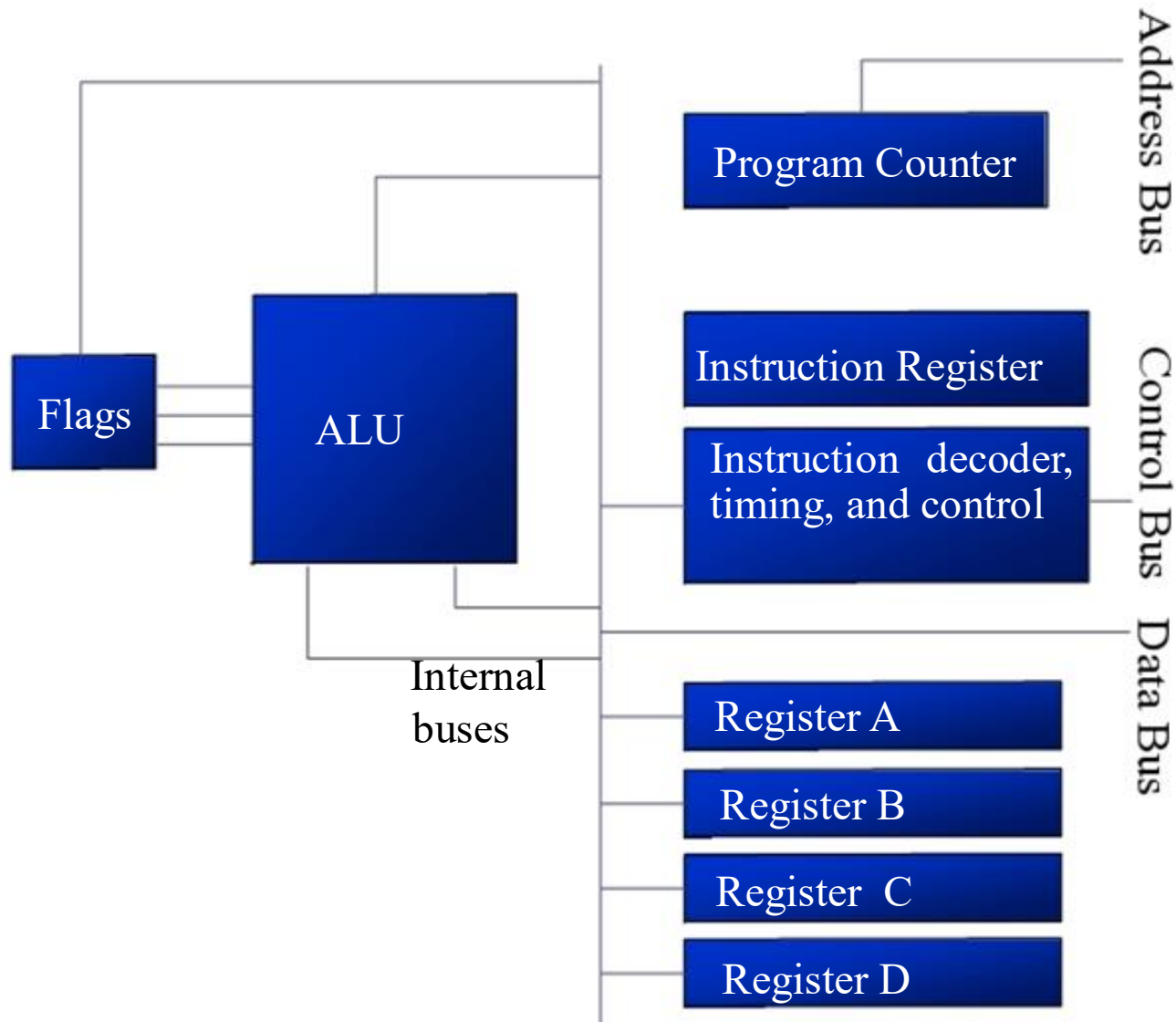
- The CPU uses registers to store information temporarily
 - Values to be processed
 - Address of value to be fetched from memory
- In general, the more and bigger the registers, the better the CPU
- Registers can be 8-, 16-, 32-, or 64-bit
- The disadvantage of more and bigger registers is the increased cost of such a CPU

Instruction Register



- of a CPU's control unit that stores the instruction currently being executed

INSIDE CPUs



CPU Execution Cycle

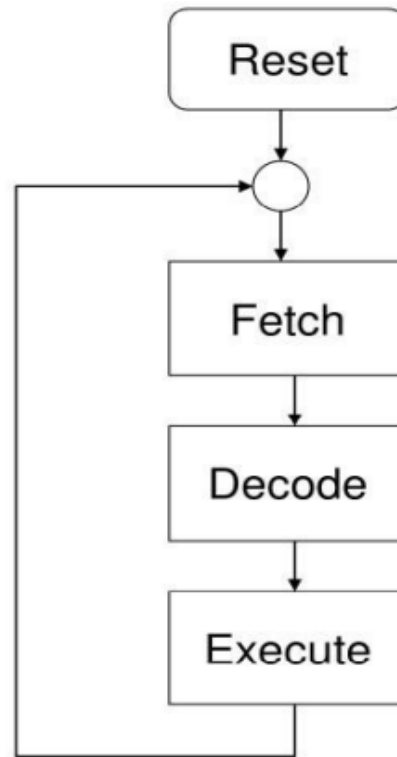


- CPU executes instructions in **endless fetch, decode, execute cycles**.
- It only knows how to do three things:
 - ☐ Fetch instructions from somewhere.
 - ☐ Analyze instruction, get more data if necessary.
 - ☐ Execute instruction.
- Keeps track of instruction using **Program Counter (PC)**:
 - ☐ Tells CPU location of next instruction.

CPU Execution Cycle



Fetch, Decode, Execute

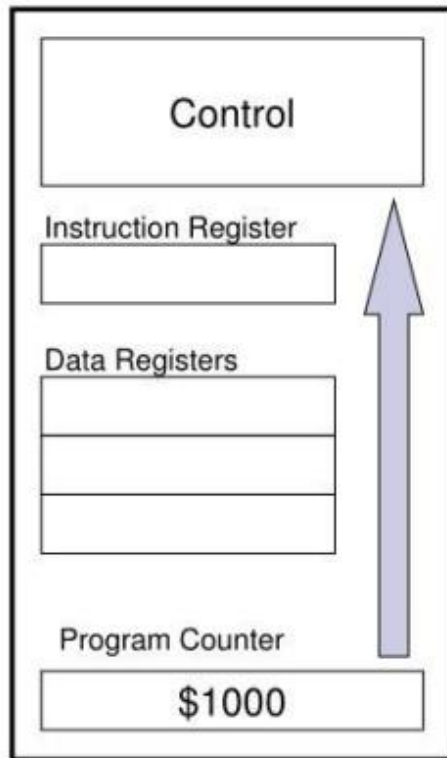


CPU Execution Cycle



Fetch – Step 1

CPU



CPU gets
instruction
address from PC

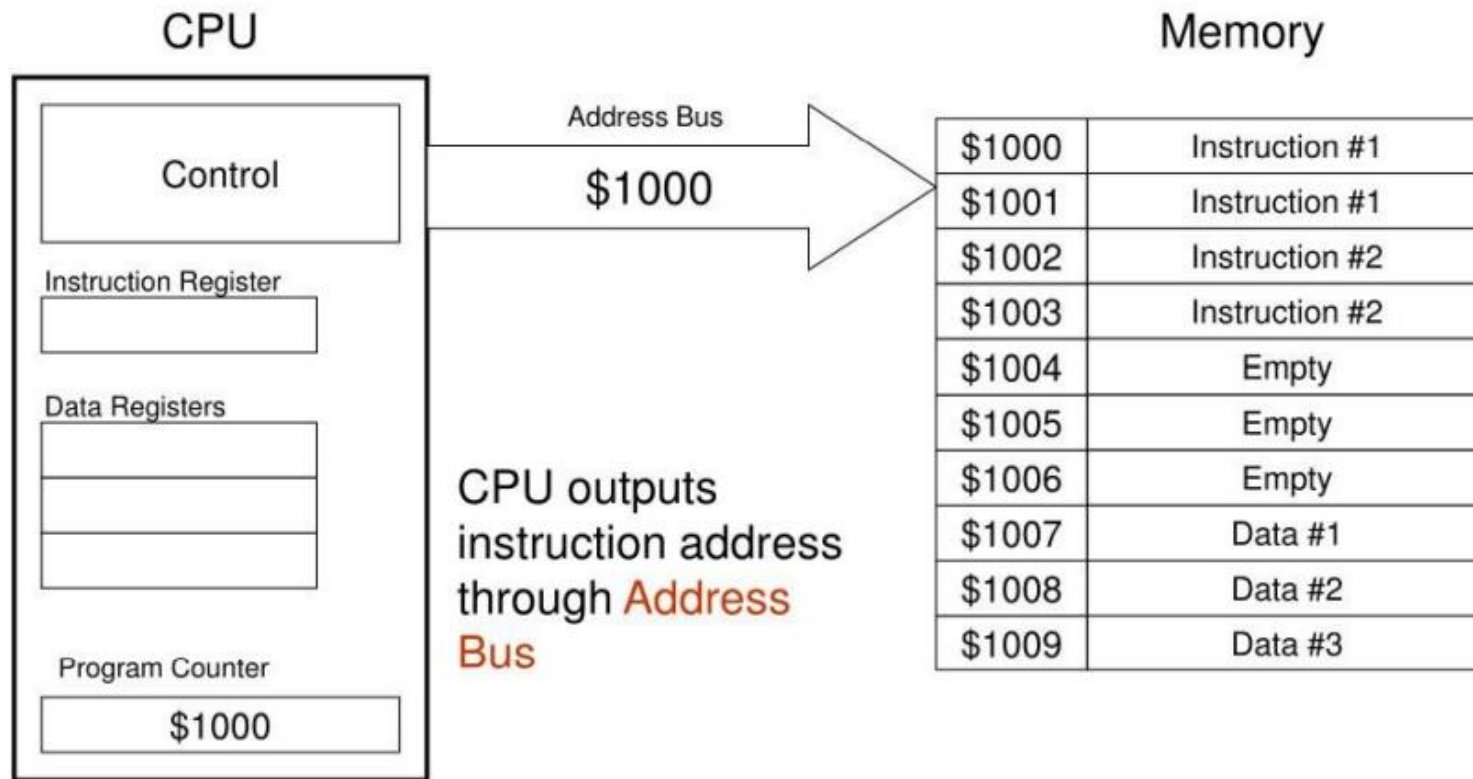
Memory

\$1000	Instruction #1
\$1001	Instruction #1
\$1002	Instruction #2
\$1003	Instruction #2
\$1004	Empty
\$1005	Empty
\$1006	Empty
\$1007	Data #1
\$1008	Data #2
\$1009	Data #3

CPU Execution Cycle



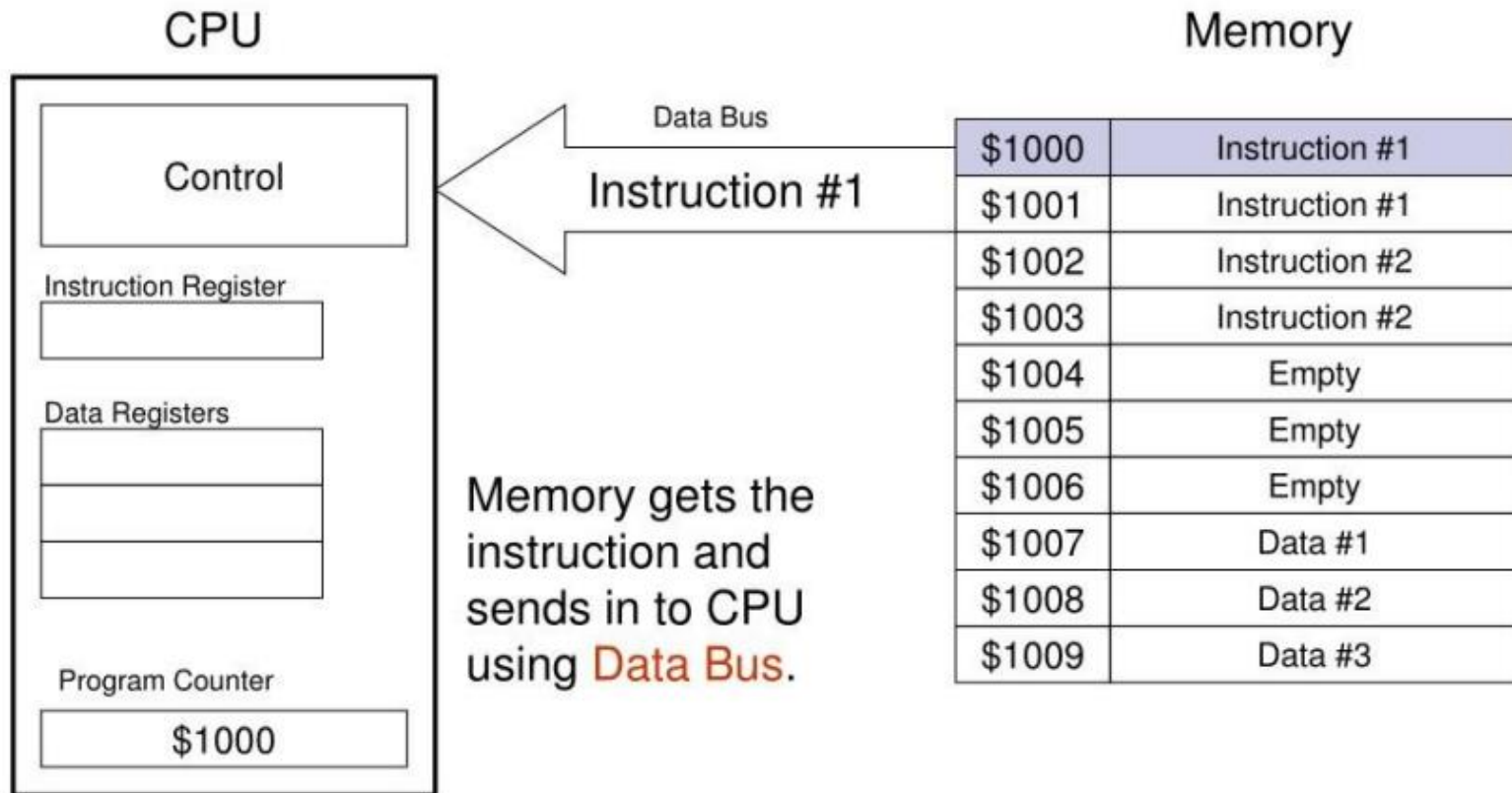
Fetch – Step 2



CPU Execution Cycle



Fetch – Step 3

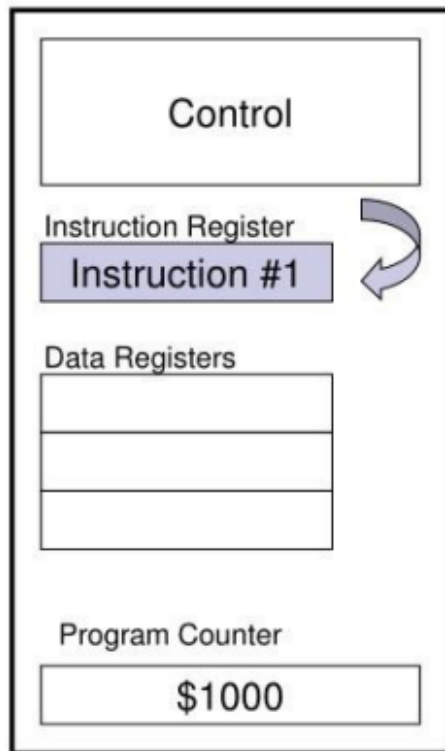


CPU Execution Cycle



Fetch – Step 4

CPU



CPU stores
instruction in
Instruction Register

Memory

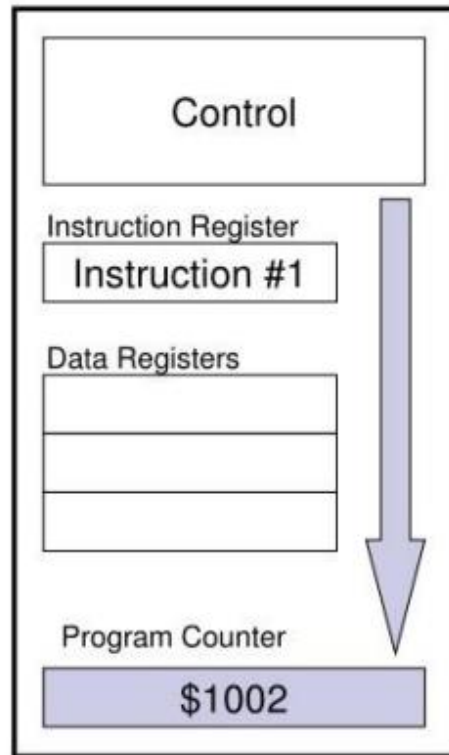
\$1000	Instruction #1
\$1001	Instruction #1
\$1002	Instruction #2
\$1003	Instruction #2
\$1004	Empty
\$1005	Empty
\$1006	Empty
\$1007	Data #1
\$1008	Data #2
\$1009	Data #3

CPU Execution Cycle



Fetch – Step 5

CPU



After instruction has been loaded, CPU updates Program Counter.

Memory

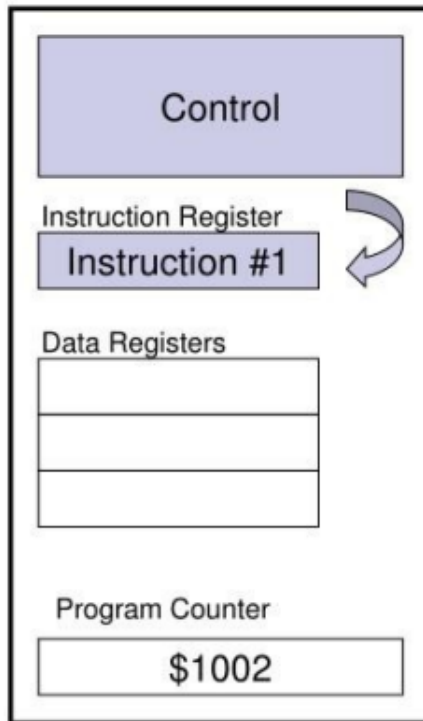
\$1000	Instruction #1
\$1001	Instruction #1
\$1002	Instruction #2
\$1003	Instruction #2
\$1004	Empty
\$1005	Empty
\$1006	Empty
\$1007	Data #1
\$1008	Data #2
\$1009	Data #3

CPU Execution Cycle



Decode – Step 1

CPU



CPU **analyzes instructions** before executing it.

Memory

\$1000	Instruction #1
\$1001	Instruction #1
\$1002	Instruction #2
\$1003	Instruction #2
\$1004	Empty
\$1005	Empty
\$1006	Empty
\$1007	Data #1
\$1008	Data #2
\$1009	Data #3

Type of instruction.

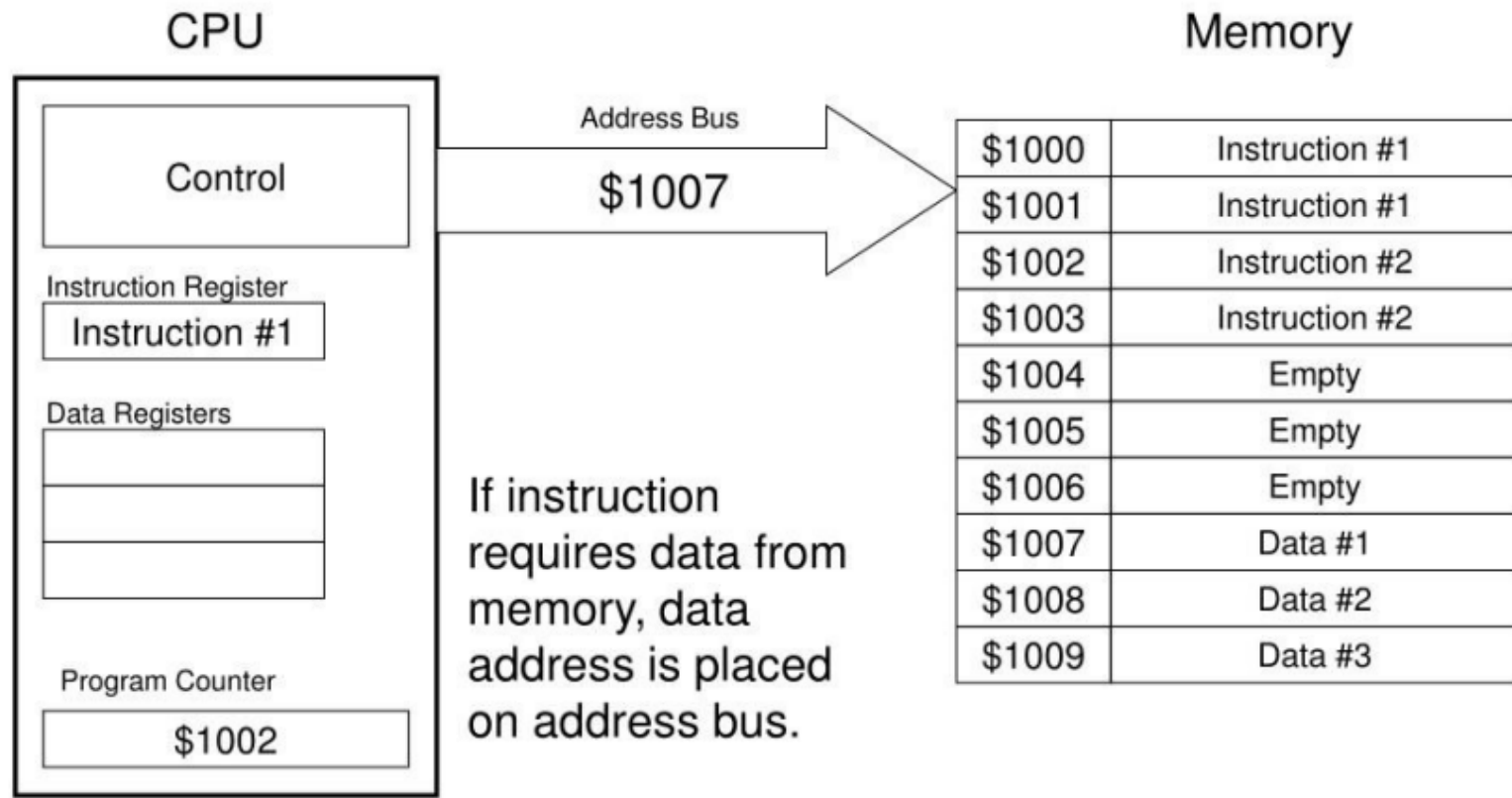
Does the instruction require any data to perform calculations?

Where are the data located?

CPU Execution Cycle



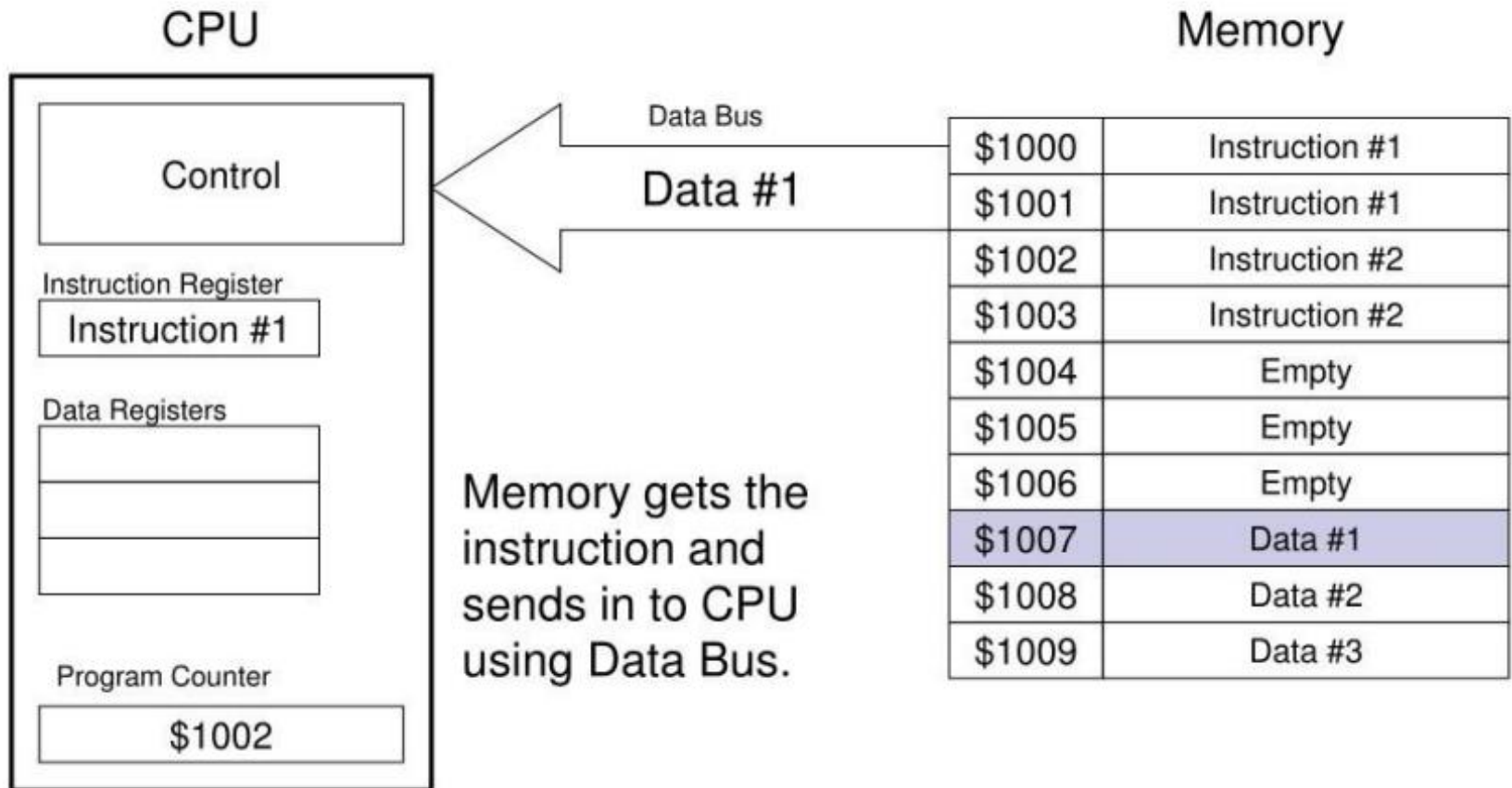
Execute – Step 1



CPU Execution Cycle



Execute – Step 2

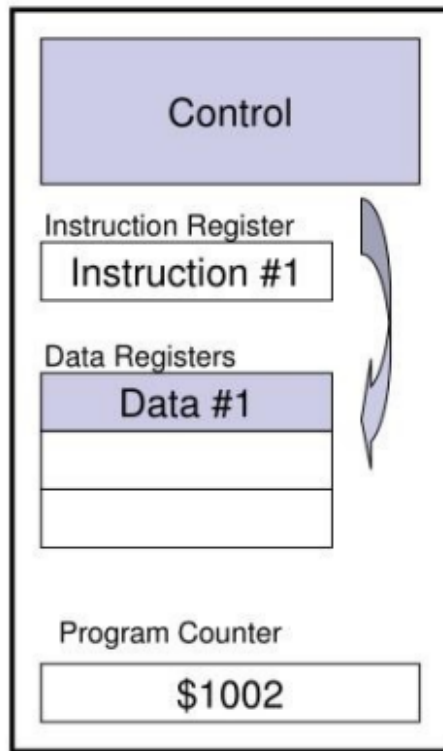


CPU Execution Cycle



Execute – Step 3

CPU



CPU puts data
inside internal data
registers and
execute instructions.

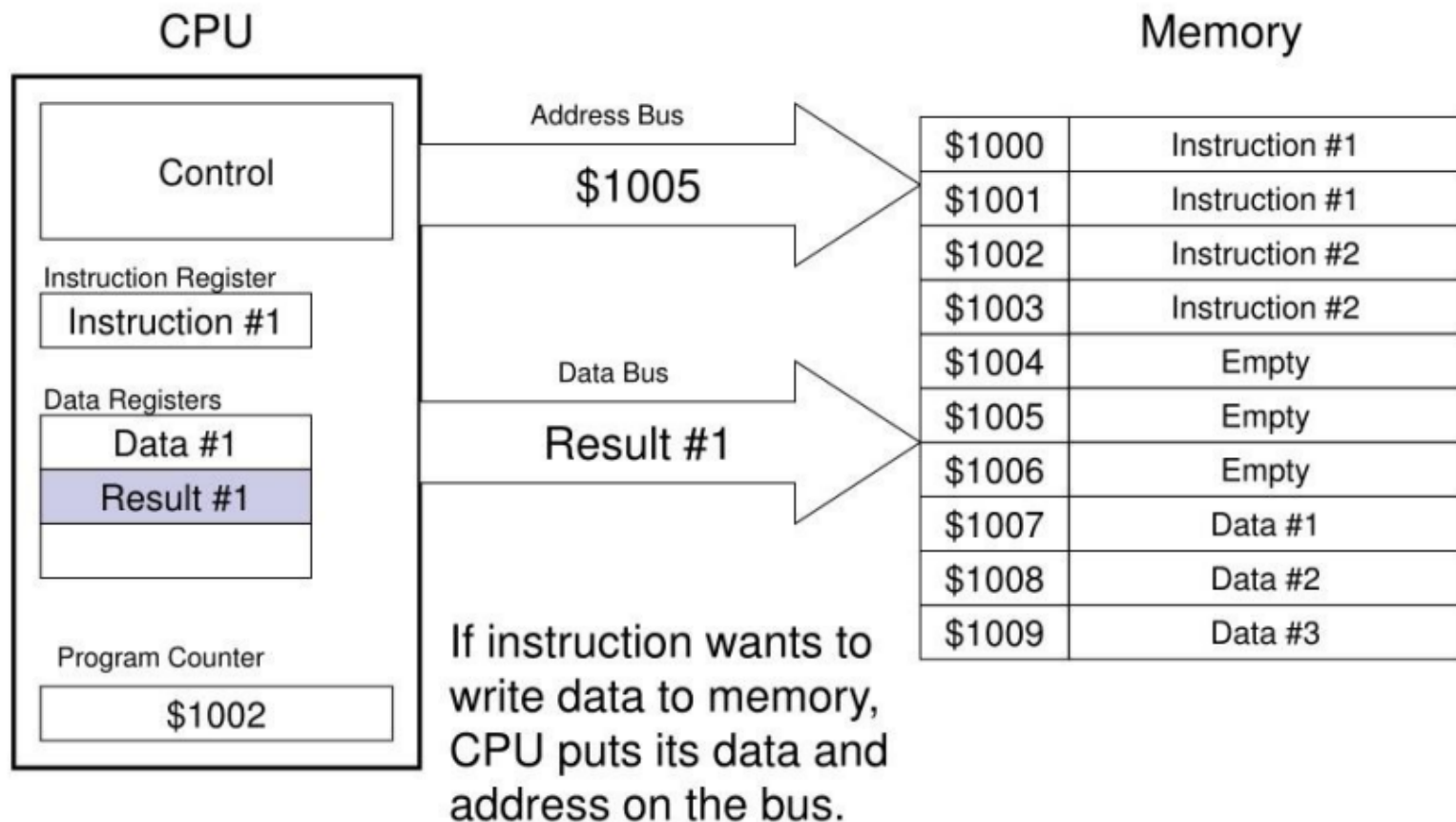
Memory

\$1000	Instruction #1
\$1001	Instruction #1
\$1002	Instruction #2
\$1003	Instruction #2
\$1004	Empty
\$1005	Empty
\$1006	Empty
\$1007	Data #1
\$1008	Data #2
\$1009	Data #3

CPU Execution Cycle



Execute – Step 4

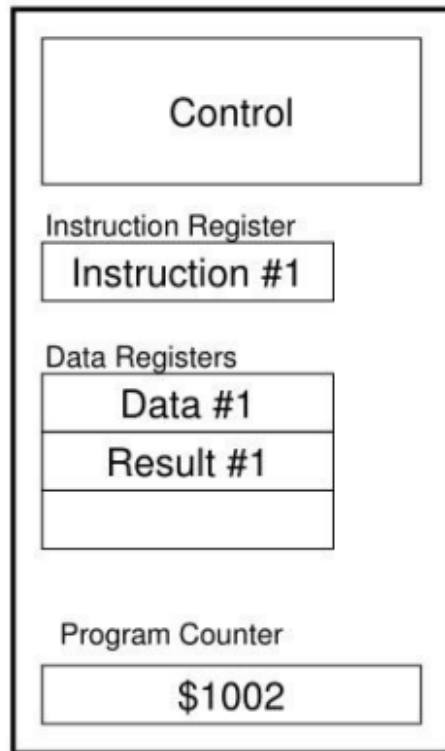


CPU Execution Cycle



Execute – Step 5

CPU



Memory

\$1000	Instruction #1
\$1001	Instruction #1
\$1002	Instruction #2
\$1003	Instruction #2
\$1004	Empty
\$1005	Result #1
\$1006	Empty
\$1007	Data #1
\$1008	Data #2
\$1009	Data #3

Memory receives instructions and puts data in the location.

INSIDE CPUs



- ALU (arithmetic/logic unit)

Performs arithmetic functions such as add, subtract, multiply, and divide, and logic functions such as AND, OR, and NOT

- Program counter

Points to the address of the next instruction to be executed

As each instruction is executed, the program counter is incremented to point to the address of the next instruction to be executed

Instruction decoder

Interprets the instruction fetched into the CPU

A CPU capable of understanding more instructions requires more transistors to design

Memories

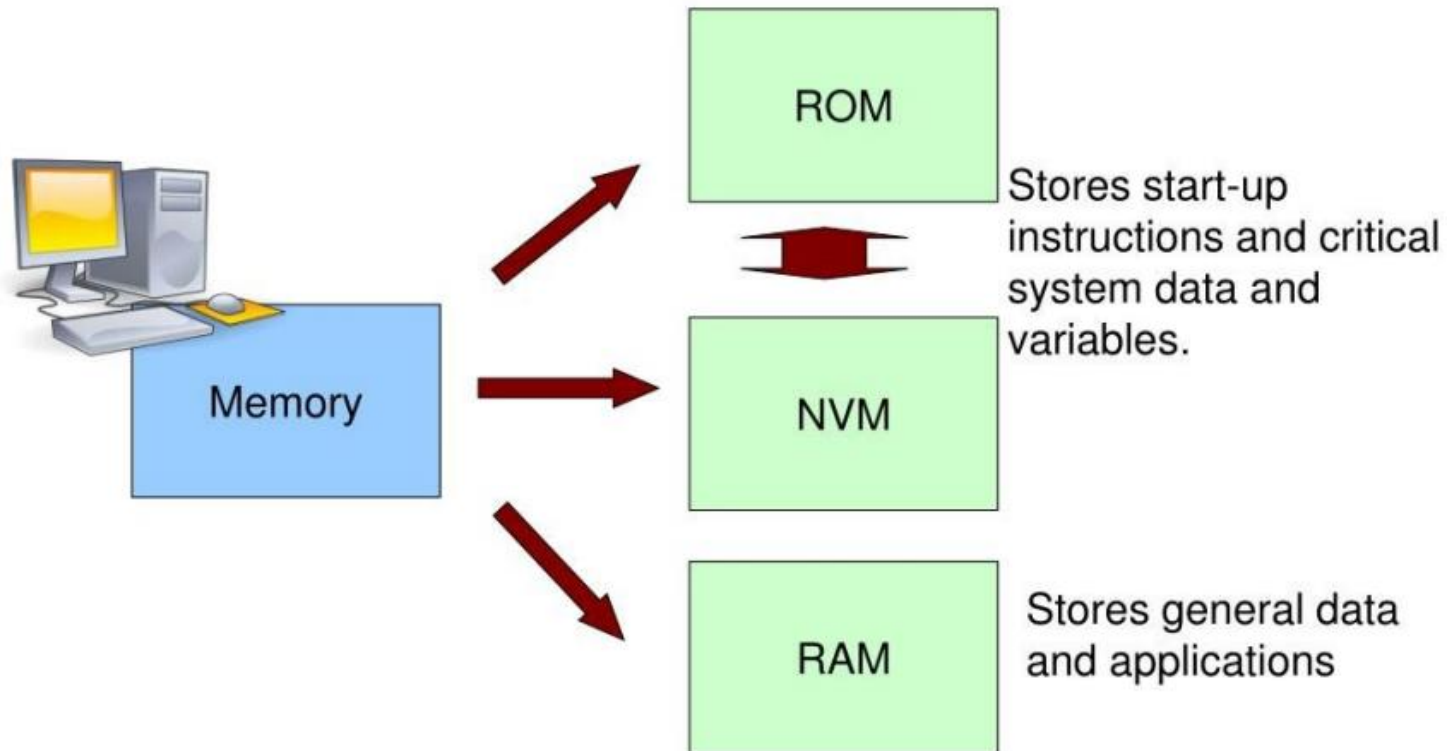


- Stores instructions and data for CPU.
- Each memory location given unique address.
 - CPU refers to address to access.
- Types:
 - Read-Only Memory (ROM).
 - Random-Access Memory (RAM).
 - Non-Volatile Memory (NVM).

Memories



RAM, ROM and NVM



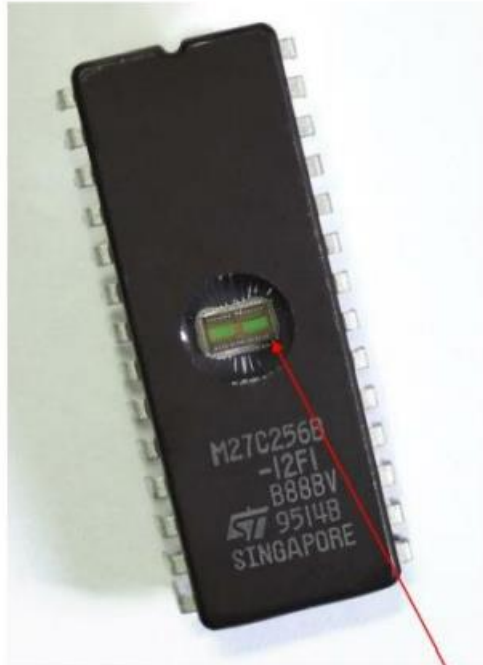
ROM



- Read-Only Memory:

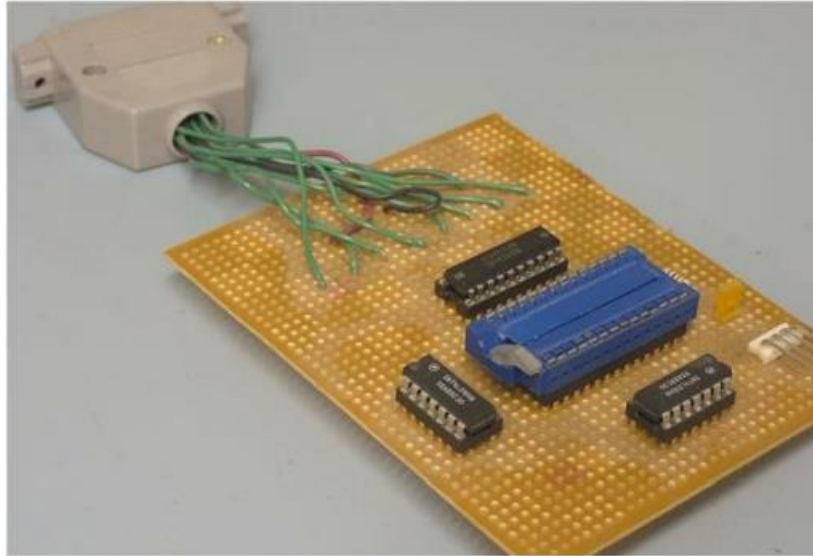
- ☐ Data **can be read**, but **cannot be written (read-only)**.
- ☐ **Contents stay** without power (non-volatile).
- ☐ Usually contains basic start-up instructions, data.
- ☐ Contents **hard-wired** during manufacturing.
- ☐ Newer versions can be reprogrammed:
 - PROM: Fuse & anti-fuse.
 - EPROM: UV light.
 - EEPROM: Electrical current.

ROM Examples



EPROM

Quartz Window



EEPROM Programmer

NVM



- Non-Volatile Memory
- Contents can be **read** and **written**.
- **Contents stay** without power (**non-volatile**).
- Advantages:
 - ☐ Keeps memory even with no power.
 - ☐ Data is protected against blackouts.
 - ☐ Rewriteable.
- Disadvantages:
 - ☐ Slower than RAM.

RAM

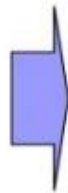


- Random Access Memory.
- Contents can be **read** and **written**.
- **Loses data** without electrical power (**volatile**).
- Advantages:
 - ☐ Programs can be loaded and reloaded.
 - ☐ Larger capacity.
- Disadvantages:
 - ☐ Requires power, refresh cycles.

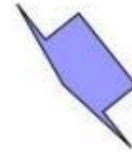
RAM vs ROM



Computer is turned on



CPU looks for instructions from memory

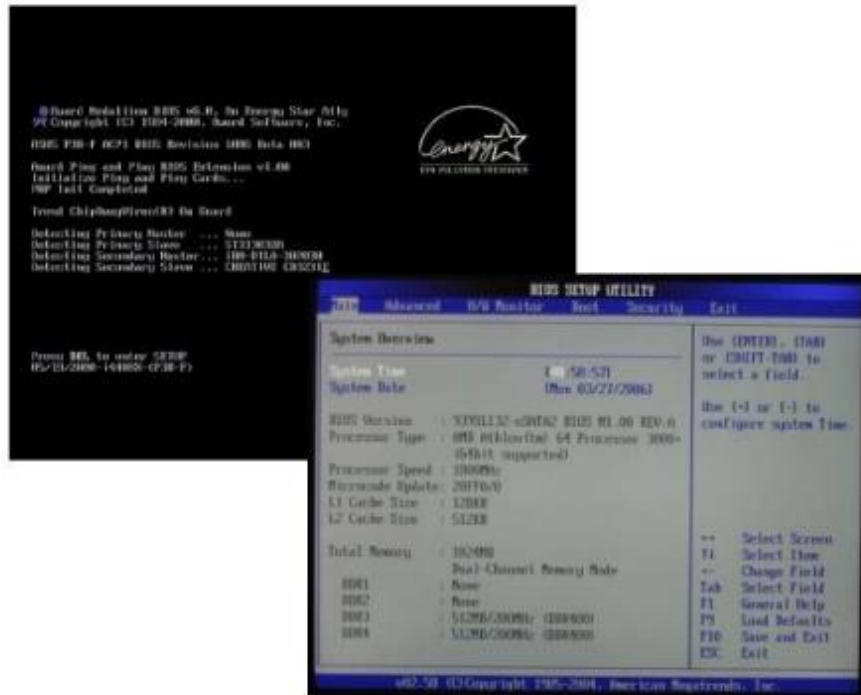


RAM is still empty because the computer has just been started.



CPU loads instructions from ROM.

RAM vs ROM



ROM only has basic functions to start the computer.



RAM loads more advanced functions, such as the OS.



- **Accessing memory**

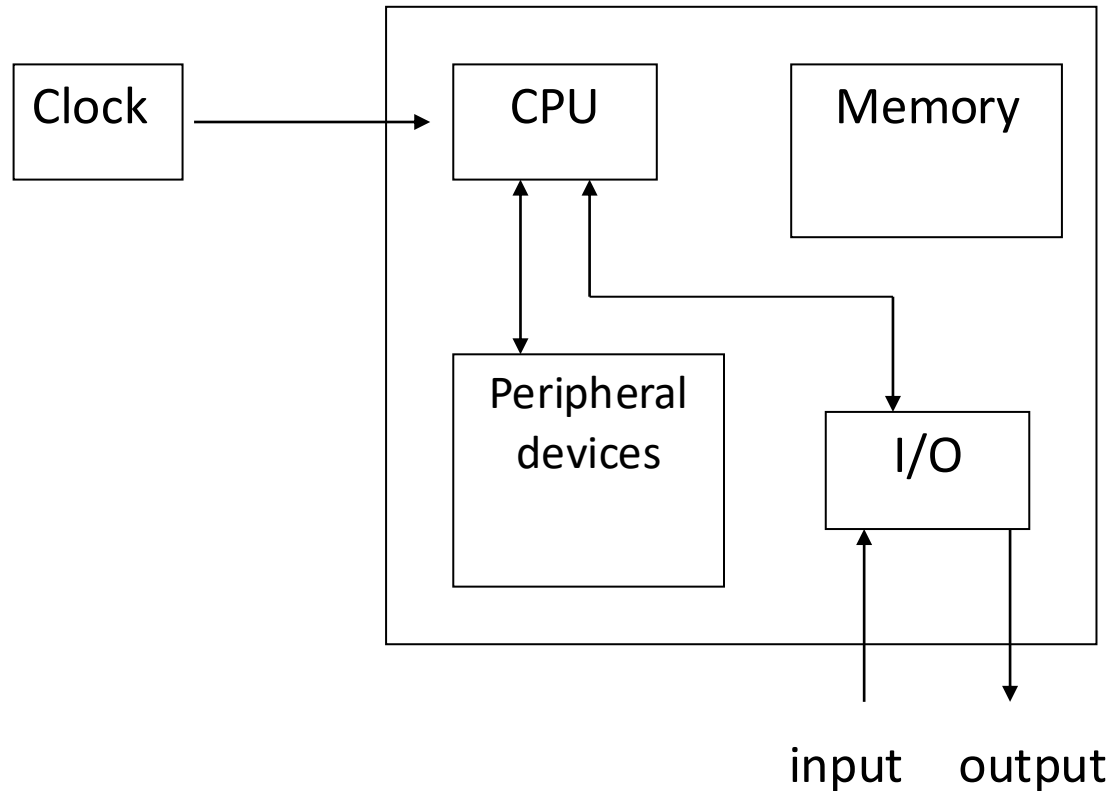
- Each location in a memory is given a number, called an address
- the 16 locations of memory would be numbered from 0 to 15, or in binary 0000–1111₂.
- The maximum number of locations that can be addressed will depend on the number of bits in the address.
- *Example:*
 - a 4-bit address can access 16 locations.,
 - 20 address lines have $2^{20} = 1\,048\,576$ or 1 Meg locations.

Memory organization



- A memory contains a number of cells or registers that, themselves store a number of bits.
- The memory organization is always quoted as '*number of locations x bits stored in each*' so this memory would have an organization of anywhere between 16 x 1, 16 x 4 or 16 x 8.
- Number of locations is equal to 2^n where n is the number of address lines.
- For example, the Pentium microprocessor uses 32-bit addresses for accessing memory words. This provides a maximum of $2^{32} = 4,294,964,296 = 4$ GB of memory addresses, ranging from 00000000, to FFFFFFFF, in hexadecimal

MICROCONTROLLER



Microcontroller



- It is simply a computer on a chip. It is entirely self-contained computers built on a single piece of silicon that incorporate all of the functions of a traditional Computer.
- It is a special device generally found in a stand alone system wherein medium to large scale computing is required. The microcontroller serve as the data acquisition/analysis and processing device in such applications
- Its name was derived from the words:
 - *Micro* – which means miniature
 - *Controller* – which is defined as a circuit that manipulates the behavior of another device or a system

Advantages of Microcontrollers



- microcontrollers are inexpensive allowing students to build and create individual, take-home projects .
- microcontrollers are inexpensive and easily replaceable in the event that they are accidentally destroyed
- microcontroller development tools are typically free or significantly discounted for promotional purposes.
- microcontroller development tools have minimal computing resource requirements (i.e.Core2 Duo, DOS or Windows 10 depending on the tools).
- microcontroller I/O ports are easily accessible in comparison with PC I/O
- microcontrollers easily interface to other electronic circuits and devices.

Disadvantages of Microcontrollers



- limited programming language and programming tool choices
- microcontrollers are well suited to simple control and interface tasks
- programs must be well-planned. Programming a device with little capability for effective feedback or debugging becomes an abstract exercise which quickly frustrates students.

Microcontroller includes:

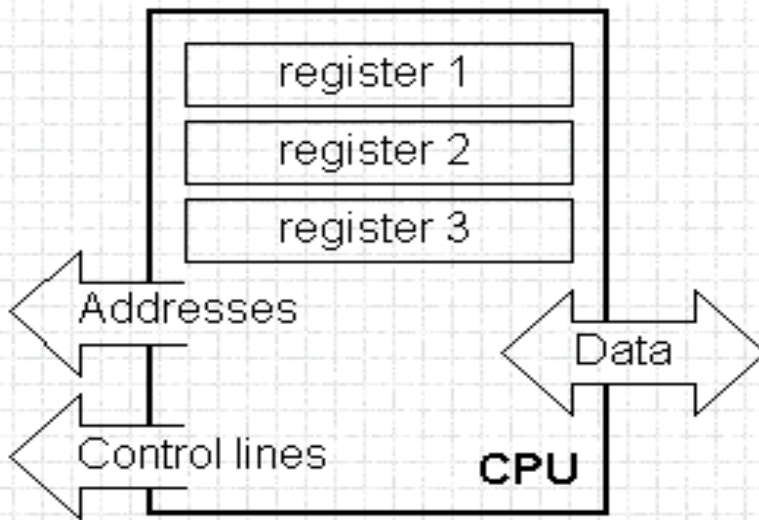


- a processing unit
- program (permanent) memory
- data (temporary) memory
- a clock/oscillator circuit
- a timer/counter circuit
- general purpose I/O ports

Central Processing Unit (CPU)



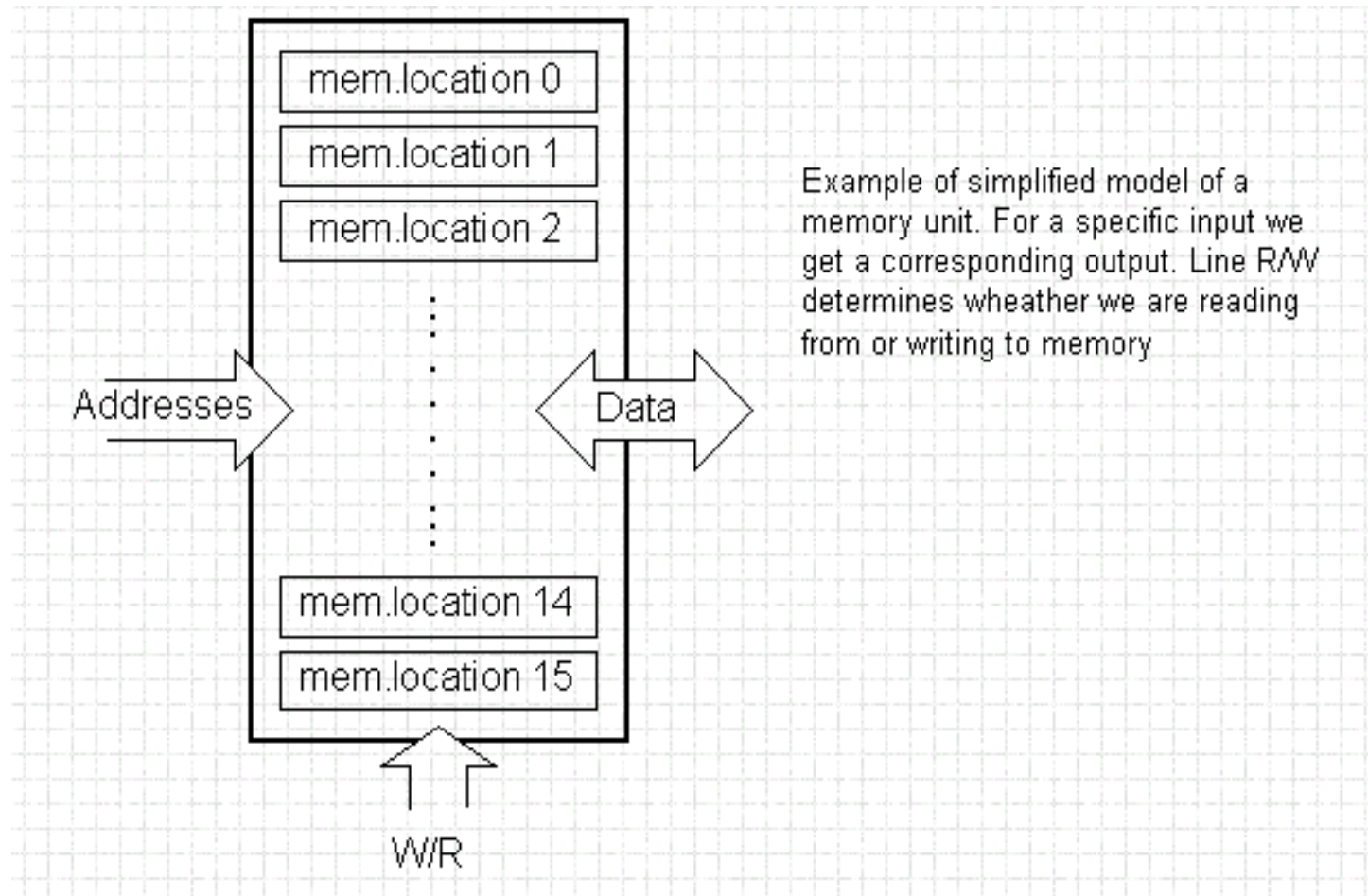
- Conduct the basic process of data analysis
- Issue commands to and from the memory units
- Produce the necessary output for each operation undertaken



Example of simplified central processing unit with three registers

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Memory Unit



Memory Chips



Random Access Memory (RAM)

- Serve as the prime storage for each data that shall and has undergone processing
- Used as the storage register for I/O functions
- Contains the CORE control systems of the microcontroller

Read Only memory (ROM)

- Serve as a secondary storage of data

A memory location wherein the firmware required for processing and analysis is written and stored. Such firmware dictates the behavior of all the incorporated devices in the microcontroller

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Peripheral Devices (Input/Output Unit)



Devices/sub-circuits primarily used to further enhance the data acquisition and transmission capability of the entire microcontroller. It allows the microcontroller to:

- Communicate with other microcontrollers units
- Communicate with other data analysis device
- Acquire both analog and digital data
- Awaken the CPU of the microcontroller when a sudden change within its port has occurred or an internal interrupt has been made (assuming the microcontroller was placed to sleep)
- Control special electric/electromechanical devices

Peripheral Devices (Input/Output Unit)



Specialized Peripheral Features such as:

- interrupt inputs
- low power and sleep modes
- A/D converters
- analog comparators
- capture/compare modules
- high current outputs
- pulse-width modulation (PWM) outputs

Pulse Width Modulation, or PWM, is a technique for getting analog results with digital means.

- serial/I2C transceivers

I2C



- also known as I2C or “eyesquared-cee”, which stands for “*inter-integrated computer communications*”
- developed originally by Philips in the late seventies as a method to provide an interface between microprocessors and peripheral devices without wiring full address, data, and control busses between devices
- also allows sharing of network resources between processors (which is known as *multimastering*)

What Are The Common Application Areas Of Microcontrollers?



Industry Application

- Logging Functions
- Control Functions
- Timing Functions
- Display Functions
- Sensor Information Processing

Household Application

- Microwaves
- Televisions
- Television remote control units

Dominant Microcontroller Brands



- Motorola
- Siemens
- ZILOG
- ATMEL
- Microchip